

Volumetric nonlinear ortho full-waveform stacking in airborne LiDAR bathymetry for reliable water bottom point detection in shallow waters

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ABSTRACT

Airborne LiDAR bathymetry allows an efficient and area-wide measurement of the water bottom topography in shallow waters. However, the maximum water depth range of this method is mainly limited by water turbidity, resulting in a reduced coverage of the water bottom topography in deeper waters. Water turbidity causes attenuation effects and hampers the reliable detection of water bottom echoes in the digitized full-waveform signal, and consequently, greater water depths are not analyzable by using standard processing methods. To increase the analyzable water depth, an extended full-waveform processing method was developed with the goal of enhancing the reliable extraction and detection of bottom points in deeper waters. This volumetric nonlinear ortho full-waveform stacking approach is based on the combined analysis of information from closely adjacent measurements under the assumption that the water depth is locally steady. Such an approach has the advantage that the influence of sensor noise and erratic non-bottom object echoes is significantly reduced so that weak water bottom echoes are better detectable. The results of the combined analysis were finally applied for water bottom echo detection in individual measurements, thus avoiding smoothing effects. For evaluation purposes, the detected water bottom points were compared with points derived from the standard processing method and with echo sounder measurements. The results of a pilot study in a river with high turbidity showed that the application of the extended full-waveform processing indicate an increase of the analyzable water depth from approximately 1.65 m to approximately 2.20 m, leading to an approximately 210% increase in the number of detected water bottom points (related to the number of standard processing) and an improved coverage of the water bottom by newly processed points.

1. Introduction

The submerged topography of inland waters and coastal zones is subject to permanent dynamic changes, which are partly caused by natural factors and partly by human intervention. Natural changes in the topography of water bodies are a result of erosion and sedimentation caused by normal flow processes or natural disasters such as floods or storm surges (Charlton, 2007; Benito and Hudson, 2010). Human-made changes result from water construction measures such as river regulation or dredging of navigation channels (Ratick et al., 1992; Bian et al., 2022) as well as coastal protection measures (Christiansen, 2016, 2021) or marine gravel and sand extraction (De Groot, 1986; Kubicki et al., 2007).

The continuous monitoring of submerged topography is important since accurate information on the geometry of the water bottom is essential for safe inland shipping, maritime navigation and harbor security. Furthermore, the mapping of the water bottom provides the basis

for the planning, implementation and assessment of active waterway management measures (e.g., groyne optimization, river engineering modifications, extension of navigation channels) and environmental protection measures (e.g., coastal protection, river restoration, seminatural hydraulic engineering) as well as for the ecological investigation of benthic habitats (Collin et al., 2008; Chust et al., 2010). The European Water Framework Directive also demands permanent monitoring of the condition of inland and coastal waters (European Union, 2000).

The most common bathymetric measurement method is echo sounding, which can determine the water bottom topography at depths of 0.2 m–11,000 m (de Jong et al., 2002; Lurton, 2002). In shallow water, however, the use of echo sounders is very time-consuming and cost-intensive due to their limited swath width. LiDAR bathymetry is an alternative measurement technology for the efficient recording of the depth and topography of shallow inland and coastal waters with water depths from 0–60 m. Current LiDAR bathymetry systems

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Table 1

Overview LiDAR bathymetry concepts. References: a = Parrish et al. (2019); b = NASA-ICESat2 (2022); c = Mandlbürger (2020); d = Quadros (2013); e = RIEGL (2021); f = RIEGL RiParameter Software. Sensors: I = ICESat-2; II = Leica/AHAB HawkEye IIB; III = Optech SHOALS-1000T; IV = RIEGL VQ-820-G; V = Leica/AHAB Chiroptera; VI = RIEGL VQ-840-G.

	AGL	Laser footprint	Typical point spacing	Depth performance
Space-borne LiDAR bathymetry (photon counting LiDAR)	500 km ^(a)	17 m ^(a)	0.7 m × 3300 m ^(b, I)	1 × Secchi depth ^(a)
Airborne LiDAR deep bathymetry	500 m–1000 m ^(c)	2 m–6 m ^(III, II)	0.5 m × 0.5 m–5 m × 5 m ^(d, II, III)	3 × Secchi depth ^(c)
Airborne LiDAR shallow-bathymetry	500 m–1000 m ^(c)	<1 m ^(IV)	0.2 m × 0.2 m–1 m × 1 m ^(d, IV, V)	1.5–2.5 × Secchi depth ^(c)
UAV LiDAR shallow-bathymetry	50 m–150 m ^(c)	0.05 m–0.15 m ^(e, IV)	0.04 m × 0.04 m–0.4 m × 0.4 m ^(f, VI)	1.5–3.0 × Secchi depth ^(c)

differ in terms of sensor and platform concepts (Mandlbürger, 2020). Space-borne instruments such as ICESat-2 are suitable for nearshore bathymetric mapping in discrete profiles (Parrish et al., 2019). Airborne LiDAR bathymetry (ALB), which includes deep-bathy systems and shallow-bathy systems, allows for area-wide recording of the underwater topography. Deep-bathy systems provide a high depth penetration of up to 60 m with low spatial resolution (Mandlbürger, 2020). Shallow-bathymetric systems are designed for the simultaneous acquisition of topography and bathymetry and offer a high measurement-point density with limited penetration depth (Mandlbürger, 2020). The highest spatial resolution is achieved by shallow-bathymetric UAV LiDAR systems operated at low altitudes from manned and unmanned platforms (Mandlbürger et al., 2020, 2021). Table 1 summarizes the characteristics of the different concepts.

LiDAR bathymetry is based on the simultaneous scanning of the water surface, water column and water bottom. For this purpose, the systems emit short green laser pulses with a wavelength of $\lambda = 532$ nm and record the complete full-waveform of the backscattered signal. To derive object information, water surface echo and water bottom echo must be detected in the resulting full-waveform. Subsequently, the water bottom coordinates can be calculated from the signal travel time, propagation velocity, position and direction of the pulse emission (Philpot, 2019). In this context, the change in propagation direction and velocity at the air–water interface and further propagation-induced effects must be taken into account (Guenther and Thomas, 1983; Richter et al., 2021a).

Currently, the depth performance of LiDAR bathymetry systems is limited to 1–3 Secchi depths. The reason for this limitation is the diffuse scattering and absorption of the laser pulses within the water body, which causes a strong attenuation of the signal amplitude (Guenther, 1985; Guenther and Goodman, 1978). In the case of higher water turbidity or greater water depth, the bottom echo in the signal is therefore very weak. The detection of such weak water bottom echoes is currently a major challenge. Section 1.1 summarizes conventional methods for evaluating full-waveform data and reviews recent developments. Section 1.2 highlights the contribution of this article to the research field.

1.1. Related work

There are a variety of methods for evaluating and obtaining object information from full-waveform data. Wagner et al. (2004) and Wang et al. (2015) presented basic methods for object echo detection (threshold, peak maximum, center of gravity, and zero crossing of the second derivative). In addition, there are more complex methods for the estimation of peak position, peak amplitudes and peak sharp parameters. Gaussian decomposition methods (Wagner et al., 2006; Reitberger et al., 2009; Mallet and Bretar, 2009) are not very suitable for the analysis of ALB data due to their specific signal characteristics. However, Schwarz et al. (2017, 2019) presented approaches that allow the extraction of the water surface, water column, and water bottom from

the full-waveform signal by exponential decomposition. Further information can be derived from the detected peaks. Schwarz et al. (2019) used *isolation*, *prominence* and *amplitude* parameters to derive statements about the *significance* of a peak. The *prominence* and *isolation* parameters were adapted from the work of Kirmse and de Ferranti (2017), in which mountain peaks were detected and characterized. Höfle and Hollaus (2010) used peak width to derive information about the vertical variation or roughness of the object. All methods mentioned are based on the isolated evaluation of the full-waveforms. Without additional information on water depth, weak peaks near the signal noise cannot be reliably detected with these methods. This can be remedied by novel evaluation methods based on the intelligent use of neighborhood information. Related work has shown that a combination of closely adjacent full-waveforms using full-waveform stacking is conducive to the detection of weak echoes (Stilla et al., 2007; Plenkers et al., 2013; Roncat and Mandlbürger, 2016). In Mader et al. (2019), we presented our first approach for applying nonlinear full-waveform stacking techniques to airborne LiDAR bathymetry data. The aim of the combined evaluation of closely adjacent full-waveforms is the amplification of weak echoes due to their redundancy together with an improvement in the signal-to-noise ratio. The first step is to define a local neighborhood. Subsequently, all full-waveforms in this neighborhood are evaluated together, resulting in a mean water depth of the neighborhood. In a filtering procedure, incorrectly determined water depths can be detected and eliminated. In the last step, the mean water depth of the neighborhood is used for the reliable extraction of the water bottom echo in each individual full-waveform. In Mader et al. (2021), an extension of the approach was presented, which contains several improvements regarding full-waveform analysis and filtering. Table 2 (Columns 2 and 3) provides an overview of the methods implemented in Mader et al. (2019, 2021), in which the analyzable water depth could be increased by 27 % in a concrete example. The results are very promising and show a significant increase in depth performance. However, the approach is limited by the following aspects:

- (1) Methodology is based on the assumption of a horizontal water surface, which has only limited validity for coastal waters;
- (2) neighborhood definition is performed at the water surface, e.g., the direction of the laser beam (Palmer scan pattern and overlapping flight strips) is neglected; and
- (3) filter parameters are determined interactively.

Limitations due to neighborhood definition can be clearly reduced by using voxel space representations. The research of Hosoi et al. (2013), Wang et al. (2007), and Richter and Maas (2022) shows the great potential of regular voxel structure as a data structure for volumetric analysis methods and data representation.

Pan et al. (2016) proposed an approach for deriving water information from full-waveform data using image-processing methods. Since the slanted ray directions are problematic for this approach, the full-waveform data are transformed into a voxel structure, with vertical voxel columns referred to as orthorectified waveforms (*orthowaveforms*). The voxel structure is analyzed using support

Table 2
Overview of previous work and current article.

	Mader et al. (2019)	Mader et al. (2021)	Current article
Neighborhood definition			
Regular raster grid	✓	✓	×
Regular voxel grid	×	×	✓
Full-waveform stacking			
Accumulation of individual full-waveforms	✓	✓	×
Volumetric stacking (ortho full-waveform)	×	×	✓
Full-waveform analysis			
Peak maxima	✓	✓	✓
Peak significance	×	✓	✓
Determination of the mean water depth			
Directly in stacked full-waveform	✓	✓	×
From derived water surface and water bottom model	×	×	✓
Filtering of false detections			
Neighboring cell check from the shore	✓	×	×
Neighboring cell check from reference water depth	×	✓	×
Analysis of full-waveform characteristics	×	×	✓
Water surface model			
Horizontal water surface	✓	✓	×
Curved water surface	×	×	✓

vector regression image-processing techniques to derive bathymetry and turbidity information. Support vector regression requires data-driven machine learning techniques and thus training datasets for its application.

The limitation due to the horizontal water surface can be overcome by introducing more complex water surface models. In Richter et al. (2021a), different modeling strategies and their effect on the accuracy of the water bottom points were discussed. A suitable modeling approach was presented in Kraus (2000) with the terrain model with composite linear bivariate polynomials or the finite element method, which is similar to the locally tilted water surface model in Richter et al. (2021a). This becomes more relevant as newer ALB sensor systems operate with high pulse rates and smaller beam divergences to achieve a high lateral resolution. Older systems used larger beam divergences with the assumption that the wave effects within the footprint are averaged out ('passive wave correction', Guenther, 1985; Westfeld et al., 2017). This results in a lower measurement resolution, but this was not important due to the low pulse rates.

1.2. Contribution

As described earlier, turbidity is a significant factor in the attenuation of the bottom echo, impeding the reliable detection of weak bottom echoes. This leads to the focus of this article: How can bottom points reliably be extracted from full-waveforms with weak water bottom echoes overcoming the limitations of the approach of Mader et al. (2019, 2021)? This article proposes an approach for processing ALB full-waveforms in shallow water with the following objectives:

- reliable extraction of water bottom information from weak echoes (i.e., increase in the analyzable water depth);
- avoiding smoothing effects and a reduction in measurement resolution;
- optimization of the extraction of the water bottom from existing datasets (without hardware-specific adaptations); and
- high reliability of the results due to suitable filtering methods.

The approach to achieve the aims formulated above is termed volumetric full-waveform stacking processing (VFWSP). It makes additional water depth information accessible by a combined analysis of closely neighboring ALB measurement data. The most important difference from previous studies by the authors is the geometric consideration of the propagation direction of the laser pulse throughout the water column. For this purpose, the measured ALB full-waveforms are transferred into a voxel space. Then, similar to Pan et al. (2016), a pseudo

full-waveform (ortho full-waveform) is generated from the vertical voxel columns. This pseudo full-waveform, in turn, is also similar to the stacked full-waveform presented in Mader et al. (2019, 2021); thus, the full-waveform analysis and filtering tools from previous work are applicable for deriving initial water depths. This information is used to detect water bottom echoes in the individual full-waveforms. The proposed processing method should reduce smoothing (low-pass filter effects) and prevent a reduction in the original measurement resolution.

This article is structured in six sections beginning with an introduction in Section 1. Section 2 presents the shallow-water LiDAR bathymetry dataset used in the study, the available reference data, and the evaluation methods. Section 3 provides a detailed description of the ortho full-waveform stacking methodology. The results of the investigation are presented in Section 4. Section 5 discusses the results and their accuracy and reliability. The article is completed by a conclusion and future work in Section 6.

2. Datasets and evaluation methods

The methods presented in this article require practical validation to assess and evaluate their functionality, accuracy, reliability and potential. For this, on the one hand, a suitable data basis and, on the other hand, appropriate validation methods and reference data are needed. For our investigations, measurement data from the Elbe River (Germany) were available. Raw data, the results of the standard LiDAR processing, and echo sounder data for evaluation purposes were provided by the German Federal Waterways and Shipping Office Elbe (WSA) and the German Federal Institute of Hydrology (BfG). The study area, the characteristics of measurement data and the echo sounder data are presented in Sections 2.1–2.3. The validation methods are described in Section 2.4.

2.1. Study area

The study area on the Elbe River is located between the German towns of Klöden and Elster. For the development and evaluation of the new methods, ALB measurements of a section of the Elbe River (Fig. 1) with a length of approximately 735 m and a varying width of approximately 88 m to 168 m were used. The same study area was already processed in Mader et al. (2021) with nonlinear full-waveform stacking, i.e., the results of the current article are comparable.



Fig. 1. The study area (part of the Elbe River). The red area is the investigation area in this article. (Map data ©2021 Google). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

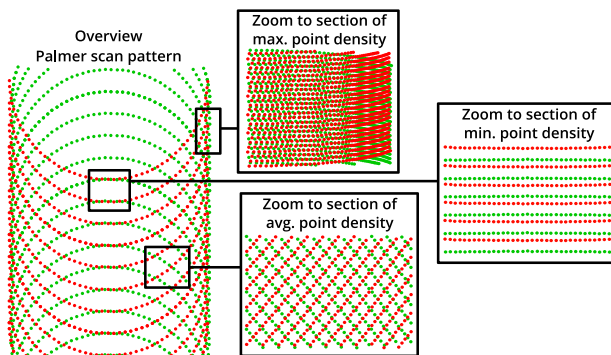


Fig. 2. Schematic Palmer scan pattern (adapted from RIEGL RiParameter software) showing the heterogeneous point density. Green points = forward look and red points = backward look. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

2.2. Measurement data

The ALB data were acquired in spring 2015 using the RIEGL VQ-880-G ALB system (RIEGL, 2018). The area was scanned at a mean flying altitude of approximately 380 m above ground and a mean flying speed of approximately 155 km h^{-1} using a Palmer scan pattern (Fig. 2) with 20° opening angle, resulting in a point density of up to 55 points m^{-2} . The used beam divergence of 1.1 mrad led to laser spot diameters on the water surface of approximately 0.44 m .

In addition to the points derived from online waveform processing (OWP; Pfennigbauer and Ullrich, 2010), the corresponding full-waveform data were recorded and digitized. In the full-waveform, all backscatterers were mapped in units of 60–200 samples (sample blocks) with a time resolution of 0.575 ns (sample time interval). This corresponds to a 3D distance of approximately 6.5 cm in water at a light propagation speed in water of $225.000 \text{ km s}^{-1}$. The maximum water depth achieved with OWP is approximately 1.65 m (Fig. 3).

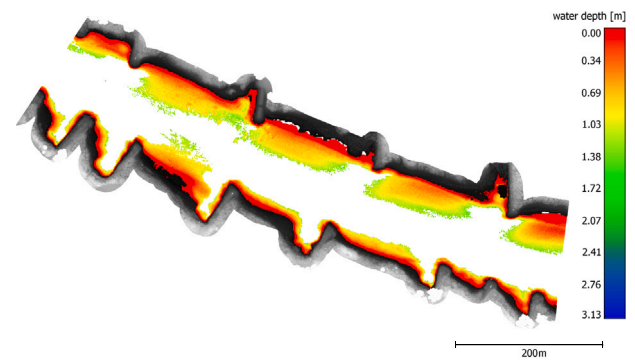


Fig. 3. The online processed data of the investigation area are color-coded according to their water depth. Land points = gray. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

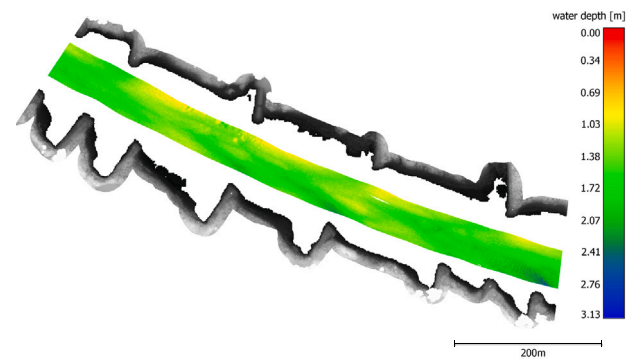


Fig. 4. The echo sounder data of the investigation area are color-coded according to their water depth. Land points = gray. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Please note that the ALB system has an optional near-infrared (NIR) channel to the green channel, but we did not have NIR data available for our developments and processing.

2.3. Echo sounder data

The echo sounder data (ES) were acquired in autumn 2015 by a measurement vessel, which was equipped with a positioning system (Leica GX1200 RTK SAPOS Rover), an orientation system (Seatex MRU5), and a vertical sweep echo sounder system (de Jong et al., 2002). The electroacoustic sounding frame consists of 37 single-beam transducers distributed on two cantilevers over a span of 12 m . The echo sounders operate with a frequency of 200 kHz and exhibit a depth resolution of 1 cm . The 3D point accuracy is stated in Weiß and Wirth (2015) with approximately 7.5 cm in embankment areas and 6.4 cm in areas with flat riverbeds. The actual maximum ES water depth measures approximately 2.7 m (Fig. 4).

2.4. Evaluation methods

For evaluation purposes, the height deviation between the water bottom points resulting from volumetric full-waveform stacking processing and those resulting from OWP and ES were examined. In addition, the model points of a water bottom model (Section 3.6) were also evaluated. The OWP data cover water depths of $0.0\text{--}1.6 \text{ m}$, while the ES data are available in water depths of $1.0\text{--}2.7 \text{ m}$. In contrast to the OWP points, which are based on the same measurement data as the VFWSP points, the ES points offer the possibility of comparison with an independent measurement method. As a consequence, the comparison with the OWP shows the similarity of the results of the established

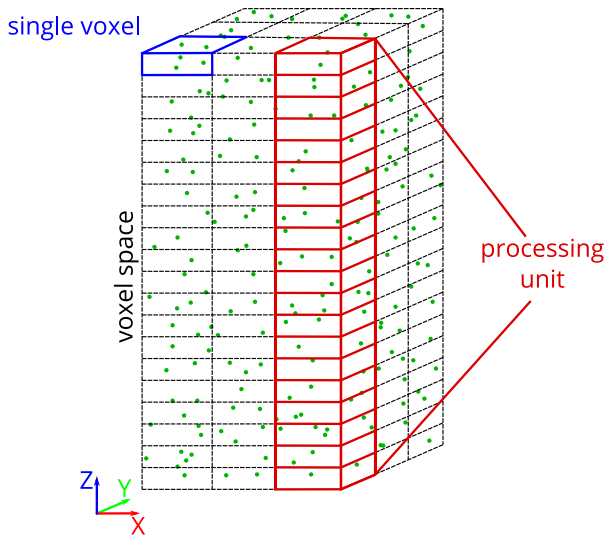


Fig. 5. Schematic illustration of a voxel space (dashed black lines), single voxel (blue), processing unit (red) and point cloud, which consists of the geo-referenced local waveform peak data (green). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

and proposed processing methods in regions where both deliver bottom points, and the comparison with the ES points validates the accuracy potential of the method.

A comprehensive evaluation of the processing results was performed by using the height deviation Δh , mean height deviation $\overline{\Delta h}$, sigma mean height deviation $\sigma_{\Delta h}$, root mean square RMS , sigma mean absolute deviation $\sigma_{MAD(mean)}$ (Hampel, 1974; Sachs, 1982), and sigma median absolute deviation $\sigma_{MAD(median)}$ (Sachs, 1982), calculated in Eqs. (1)–(6).

$$\Delta h_i = h_i(ref) - h_i \quad (1)$$

$$\overline{\Delta h} = \frac{\sum(\Delta h_i)}{N} \quad (2)$$

$$\sigma_{\Delta h} = \sqrt{\frac{\sum((\Delta h_i - \overline{\Delta h})^2)}{N}} \quad (3)$$

$$RMS = \sqrt{\frac{\sum(\Delta h_i^2)}{N}} \quad (4)$$

$$\sigma_{MAD(mean)} = 1.2533 \cdot \frac{\sum |\Delta h_i - \overline{\Delta h}|}{N} \quad (5)$$

$$\sigma_{MAD(median)} = 1.4826 \cdot \text{med}(|\Delta h_i - \text{med}(\Delta \mathbf{h})|) \quad (6)$$

with

$$i = 1 \dots N$$

N = number of height deviations

$h_i(ref)$ = height component of reference data

h_i = height component of calculated data

$\Delta \mathbf{h}$ = vector of height deviations

Δh_i = height deviation

$\overline{\Delta h}$ = mean of height deviations

$\text{med}(\mathbf{X})$ = median of a vector $\mathbf{X}$.

The RMS value and the σ_{MAD} value may differ because the RMS value contains possible systematic errors or biases. The MAD values are calculated with respect to the mean or median values. In the case of a normal distribution of the height differences, $\sigma_{MAD(mean, median)}$ is

usually used for estimating the quality of the results to avoid a too optimistic accuracy statement.

To assess the quality of the developed methods as comprehensively as possible, reliability measures were also used in addition to the accuracy analysis. For this purpose, inlier rates (Litman et al., 2015) were used, which indicate the proportion of points that do not exceed a specified limit value. As an orientation for this limit, we used the total vertical uncertainty (TVU) defined in the Special Order provided in the IHO Standards for Hydrographic Surveys (International Hydrographic Organization, 2020):

$$TVU = \sqrt{(0.25 \text{ m})^2 + (0.0075 \cdot d)^2} \quad (7)$$

with

$$d = \text{water depth in [m]}. \quad (8)$$

For the data at hand, the TVU was 0.25 m. Additionally, to obtain a better impression of the reliability of the results, the inlier rates for 0.15 m and 0.35 m are also shown. For a first overall impression of how the evaluation method works, the determination of the statistical values was initially carried out for all VFWSP points. However, the accuracies as well as the reliability measures can vary with respect to the water depth; therefore, they were also analyzed depending on the water depth.

3. Methods

This section addresses the volumetric nonlinear ortho full-waveform stacking processing and its methods for the detection and extraction of water bottom points from ALB full-waveform data. The basic idea of full-waveform stacking processing is a joint evaluation and analysis of closely neighboring measurement data to realize reliable detection and extraction of weakly pronounced water bottom echoes in full-waveform data. The method is based on the assumption that adjacent measurement data have similar properties with respect to the water surface echo and the water bottom echo. The combination of closely neighboring measurements is realized by transforming the measurement data into a voxel space. A prerequisite for this is the modeling of the water surface (both height and slope) needed for refraction correction (Section 3.1). First, all full-waveform information is mapped into a voxel space (voxelization; Section 3.2). Then, vertically stacked voxels are combined into a vertical voxel column (vertical water column), which corresponds to a processing unit (Fig. 5). Subsequently, the water column information is extracted into a signal (very similar to a full-waveform), which has advantageous properties for subsequent processing, namely, an improved signal-to-noise ratio and a strictly vertical direction (Sections 3.3). This signal is called the ortho full-waveform in the following. Based on an ortho full-waveform analysis (Sections 3.4–3.6), search corridors for the detection of water bottom echoes in the individual full-waveforms can be derived (Section 3.7). In this way, water bottom points can be extracted that were not considered in the standard processing. In Fig. 6, an overview of the presented methods, their results and the corresponding section in the methods section are given. The workflow illustration starts with the processing of the ALB data and does not include preprocessing steps such as data import, online waveform processing, or classification. The individual methods will be presented in the following section in the order of the workflow.

3.1. Water surface modeling

The volumetric full-waveform stacking approach aims to generate results with the highest possible accuracy and reliability. To achieve an accurate geometric representation of the water bottom, the refraction and change in propagation speed according to Snell's law must be considered in ALB data processing. Therefore, the following three aspects are important: (1) the precise position of the water surface, (2)

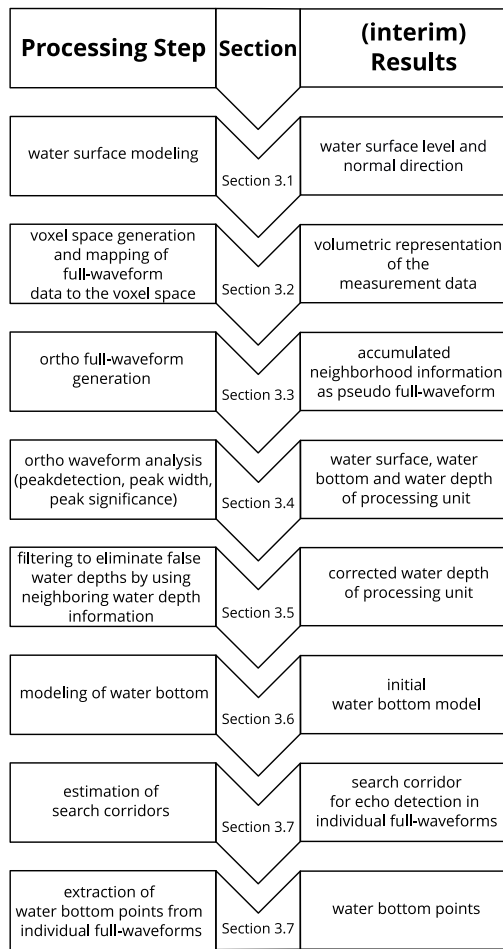


Fig. 6. Workflow of volumetric nonlinear full-waveform stacking processing. On the left side are the methods, in the middle are the corresponding sections, and on the right are the interim results or, in the last case, the final result.

an accurate determination of the water surface normals and (3) an exact definition of the search corridor for the water bottom echo detection in the individual full-waveform, including the information of neighboring processing units.

The water surface and water bottom are basically represented by XYZ -points (irregularly distributed measurement points or gridded result points). However, the representation by points is not sufficient for the above purposes; therefore, description of a continuous surface becomes necessary. For this purpose, the terrain model with composite linear bivariate polynomials or the finite element method (FEM) (Fig. 7) from Kraus (2000) was used for both water surface and water bottom modeling (Section 3.6). The basis is a regular lateral grid. The vertices NP_i of the grid cells are nodes or nodal points with lateral X - Y -coordinates defined by the regular grid and Z -coordinates to be determined in an adjustment by using Eq. (9).

$$r_s = \left(1 - \frac{\bar{X}_s}{\Delta XY} - \frac{\bar{Y}_s}{\Delta XY} + \frac{\bar{X}_s \bar{Y}_s}{\Delta XY^2}\right) \cdot Z_1 + \left(\frac{\bar{X}_s}{\Delta XY} - \frac{\bar{X}_s \bar{Y}_s}{\Delta XY^2}\right) \cdot Z_2 + \left(\frac{\bar{Y}_s}{\Delta XY} - \frac{\bar{X}_s \bar{Y}_s}{\Delta XY^2}\right) \cdot Z_3 + \left(\frac{\bar{X}_s \bar{Y}_s}{\Delta XY^2}\right) \cdot Z_4 - Z_s \quad (9)$$

with

X_s, Y_s, Z_s = coordinates of measuring point

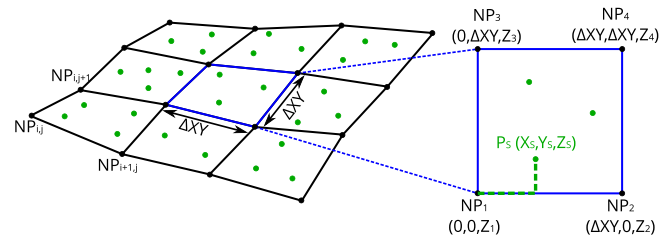


Fig. 7. Principle of the terrain model with composite linear bivariate polynomials (Kraus, 2000). The X - Y -coordinates of P_s are local coordinates related to the origin of the grid cell (NP_1).

ΔXY = grid cell width

Z_1, Z_2, Z_3, Z_4 = Z -coordinates of node points

r_s = residuum

Due to the lack of water surface height information (e.g., NIR laser scanner data or similar), the nodal points of the water surface model are determined from the OWP water surface points, which are expected to be slightly lower than the actual water surface due to a mixture of interface return and near-surface volume backscattering (Guenther et al., 2000; Mandlbürger et al., 2013). This results in an underestimation of the water surface (see Section 5.6), which can lead to inaccuracies in the water bottom estimation. The following conditional equations for the curvature in the X - and Y -directions are introduced to account for slope differences (Kraus, 2000):

$$r_{curve} = \frac{1}{\Delta XY} \cdot (Z_{j-1} - 2 \cdot Z_j + Z_{j+1}) \quad (10)$$

with

r_{curve} = residual of curve constrain

ΔXY = grid cell size

Z_{j-1}, Z_j, Z_{j+1} = Z -coordinates of consecutive node points

After estimating the nodal points, the model can be used to determine the water surface normal. For this purpose, the intersection of the laser beam (narrow laser beam model) and the water surface model is calculated by inserting Eq. (11) into Eq. (9) and solving for t .

$$\begin{pmatrix} X_{isp} \\ Y_{isp} \\ Z_{isp} \end{pmatrix} = \begin{pmatrix} X_0 \\ Y_0 \\ Z_0 \end{pmatrix} + t \cdot \begin{pmatrix} X_{dir} \\ Y_{dir} \\ Z_{dir} \end{pmatrix} \quad (11)$$

with

$X_{isp}, Y_{isp}, Z_{isp}$ = coordinates of intersection point

X_0, Y_0, Z_0 = coordinates of the origin of the laser pulse emission

$X_{dir}, Y_{dir}, Z_{dir}$ = normalized direction vector

t = distance traveled in air

Finally, the water surface normal is determined by using three auxiliary points, calculated in the radius of the laser beam spot around the intersection point (Fig. 8). Since the auxiliary points are part of the water surface, the elevation coordinates are calculated with Eq. (9) by using the corresponding nodal points. These auxiliary points can be used to determine a plane and thus also the plane normal N .

3.2. Partitioning of survey data — voxelization

The volumetric full-waveform stacking approach is based on the transformation of the laser bathymetry data into a local Cartesian voxel space representation. Thus, instead of accumulating individual full-waveforms that are only closely adjacent at the water surface into

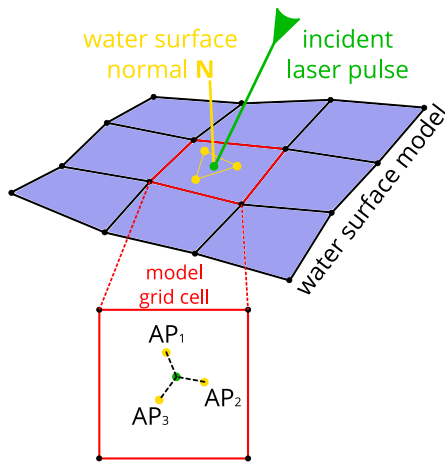


Fig. 8. Principle of water surface normal calculation. Green point = intersection point of laser beam and water surface model. Yellow points = auxiliary points (AP_1 , AP_2 , AP_3). Yellow triangle = auxiliary plane for calculating the water surface normal. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

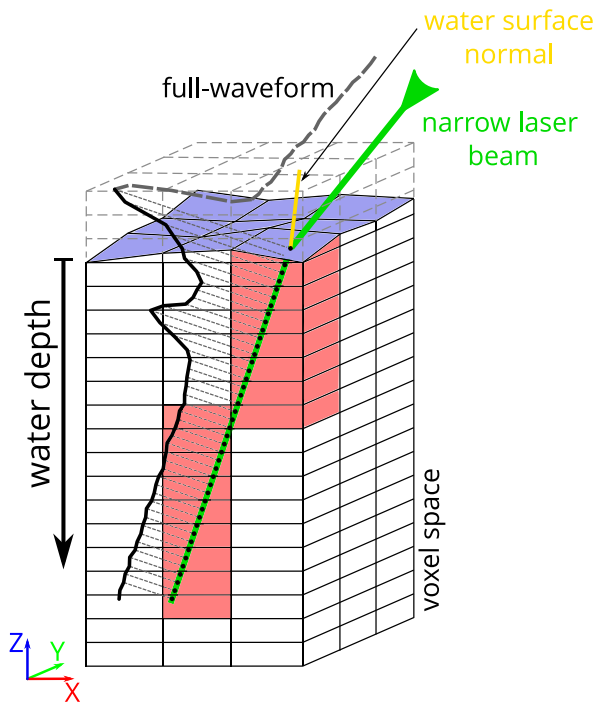


Fig. 9. Voxelization of full-waveform data: Principle of transfer from individual full-waveform samples to voxel space.

a stacked full-waveform, the individual full-waveforms are geometrically mapped into a voxel space. In detail, this means that the signal amplitudes of the geo-referenced samples of each full-waveform are transferred and resampled into the voxel space (Fig. 9). For this purpose, first, the direction vector X_1 from the point of pulse emission P_{Is} to the predetected water surface point P_{ws} is calculated (Fig. 10). Subsequently, the intersection point P_{wsm} of the vector with the water surface model (Section 3.1) and the corresponding water surface normal N are determined. At point P_{wsm} , refraction correction is applied, and the underwater direction vector X_2 is calculated (Eq. (12); Glassner, 1989).

$$X_2 = \eta \cdot X_1 + (\eta \cdot C - \sqrt{1 + \eta^2 \cdot (C^2 - 1)}) \cdot N \quad (12)$$

$$C = -N \cdot X_1$$

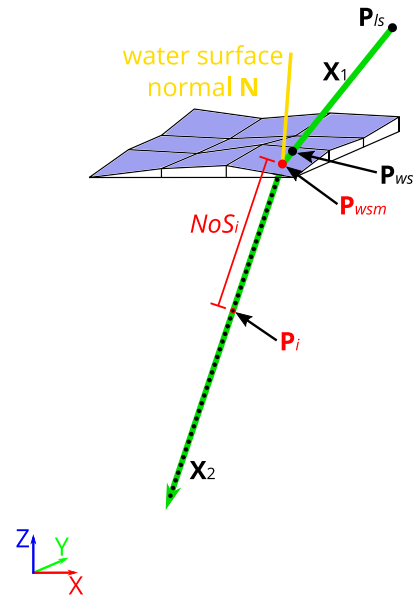


Fig. 10. Principle of geo-referencing of single full-waveform samples.

$$\eta = n_{air}/n_{water}$$

with

$$n_{air}/n_{water} = \text{refractive index air/water}$$

N = water surface normal vector

Finally, for each sample i of the full-waveform, the number of samples NoS_i (water surface echo to current sample) can be determined, and the corresponding 3D point P_i can be calculated (Eq. (13)).

$$P_i = P_{wsm} + 0.5 \cdot NoS_i \cdot t_{sti} \cdot c_w \cdot X_2 \quad (13)$$

with

$$t_{sti} = \text{sample time interval}$$

$$c_w = \text{velocity of light in water}$$

The assignment to the voxels is decided by the shortest 3D distance to the voxel center (nearest neighbor resampling). The approach may also be extended to bi/trilinear or higher-order resampling methods. If a voxel is filled with more than one sample, the signal amplitude values are averaged. Empty voxels are avoided by the choice of appropriate voxel dimensions. The divergence of the laser beam is neglected, as proper handling of dispersion and wave effects would drastically enhance complexity (Westfeld et al., 2017).

3.3. Volumetric nonlinear full-waveform stacking — generation of an ortho full-waveform

This section describes the basic methodology for evaluating neighborhood information to detect weak water bottom echoes. In volumetric full-waveform stacking, the neighborhood is defined by the vertical voxel column. To be able to use the proven full-waveform analysis methods for nonlinear waveform stacking from Mader et al. (2021), the information from the vertical voxel column is converted into a similar form as the stacked full-waveforms. For this purpose, an artificial full-waveform is created by interpreting each voxel of the processing unit as a sample of the artificial full-waveform (Fig. 11). This pseudo full-waveform is called an ortho full-waveform in the following sections and is very similar to the orthowaveform in Pan et al. (2016). The ortho full-waveform can be considered a representative signal of the processing unit. The properties of an ortho full-waveform can be summarized as follows:

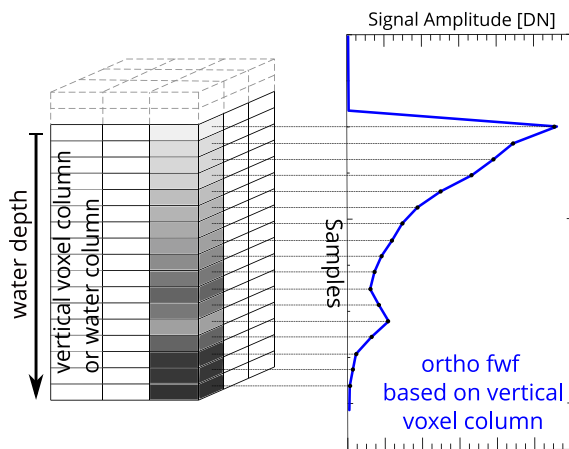


Fig. 11. Generation of an ortho full-waveform from the vertical voxel column.

- Strictly speaking, the ortho full-waveform is not a full-waveform in the true sense because it does not consist of recorded and digitized backscatter signals by a receiver (Vosselman and Maas, 2010). Consequently, the laser-radar equation (Wagner et al., 2006) cannot be applied to this signal. The ortho full-waveform is a signal based on the amplitude values of the voxel column; therefore, it is a pseudo full-waveform.
- The ortho full-waveform is composed of parts of individual full-waveforms (those that have passed the voxels of the corresponding voxel column).
- The sample distance can be simplified to the vertical voxel size.
- Since the assignment between the samples of the ortho full-waveform and the voxels in the voxel space is simple, the water depth can be determined easily.
- The methods of full-waveform analysis (Section 3.4) can be applied.

3.4. Analysis of full-waveforms

For the evaluation of the ortho full-waveform or full-waveforms in general, the methods presented in Mader et al. (2021) were employed. For the sake of completeness and to ensure a complete understanding of the methods regarding volumetric full-waveform stacking, they are briefly repeated here.

3.4.1. Peak detection

In full-waveform data, the interactions with objects are represented as echoes or local maxima (or peaks). Between two echoes, the time of the signal can be estimated (time-of-flight), which in turn is important information for the calculation of 3D points. For an accurate estimation of the peak-to-peak time, an appropriate detection of those echoes is necessary. In this article, we use the peak maximum method for the detection of water surface and water bottom echoes because it is simple to implement, has a short processing time, works reliably and has shown good accuracy in previous work (worst case scenario: half sampling time interval = approximately 0.032 m (Z-direction) in water depending on the ALB system used). Furthermore, the focus of this article is on the development and evaluation of full-waveform stacking techniques.

3.4.2. Peak significance

Each peak or local maximum of a full-waveform signal can be characterized with the parameters *isolation*, *prominence*, *amplitude* and *significance* (Fig. 12). The *isolation* describes the number of samples until the next signal part with at least the same amplitude and is a measure of how unique/special this echo is in the surrounding signal

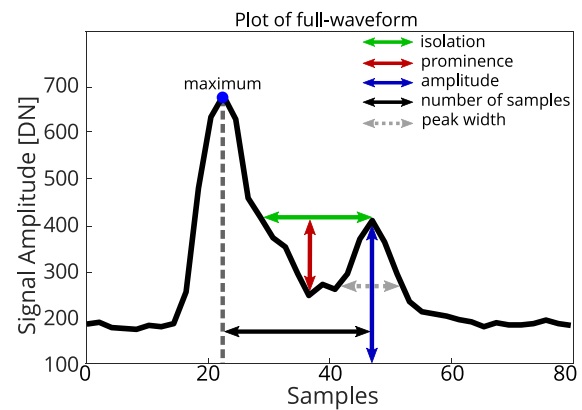


Fig. 12. Schematic representation of the parameters of a full-waveform analysis. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

part. The *prominence* is the difference between the current peak and the local minimum in the isolation area and indicates how present an echo is compared to the surrounding signal amplitudes. The *amplitude* is the received and digitized part of the backscattered signal (emitted by the sensor) on the receiver. The product of *isolation*, *prominence* and *amplitude* results in *significance*, whereby the parameters can be weighted differently if needed. With this approach, peaks of signal noise and other erratic signals can be reliably filtered in most cases. Consequently, it can be assumed that, related to ALB data, the most significant peak represents the water surface echo and the most significant peak after the water surface represents the water bottom echo, unless there is underwater vegetation.

3.4.3. Peak width

Terrain slopes and resulting different water depths have an influence on the shape of the water bottom peak when closely adjacent full-waveforms are combined. Assuming all measurements have the same water depth, this would be represented by a stronger and more narrow distinct water bottom echo in the ortho full-waveform. If changes in water bottom topography and thus water depth occur more frequently and on a larger scale, the water bottom echo becomes weaker and wider and is also more difficult to detect. This must also be considered when determining the search corridor in the individual full-waveform; otherwise, the water bottom echoes cannot be detected correctly or not at all because they are located outside the search corridor or tolerance range. To consider the height variation, the peak width is introduced and used (Fig. 12). The determination of the width of the water bottom peak for each ortho full-waveform is performed in the following way:

1. determination of the maximum of the water bottom echo;
2. determination of the first minimum between the water surface and water bottom echo starting from the water bottom echo; and
3. the number of samples between the water bottom echo (1.) and minimum (2.) is equal to half the peak width.

3.5. Filtering of volumetric full-waveform stacking results

The results of the ortho full-waveform analysis are mostly very reliable. However, false detections of water bottom echoes may occur in deeper areas where the amplitude of the water bottom echo is close to the noise level. These false detections are corrected or eliminated in an iterative filtering process. The basic idea is to check all processing units, starting from processing units with reliably determined water bottom echoes.

The decision whether a water bottom echo has been reliably detected is made by comparing its *prominence* with the noise characteristic

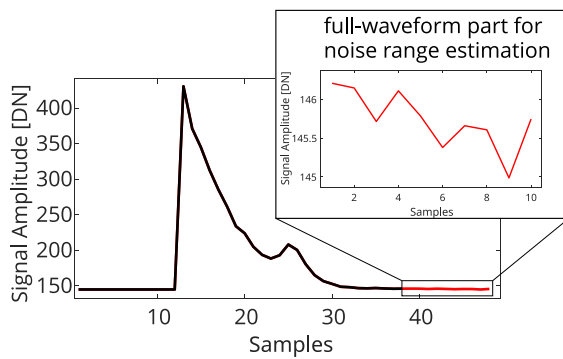


Fig. 13. Full-waveform stacking results (black) with full-waveform part for noise range estimation (red). Black curve = Complete ortho full-waveform. Red curve = part for noise range estimation. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

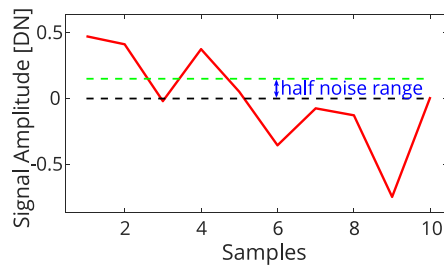


Fig. 14. Estimation of noise range. Red line = tail part of the full-waveform. Black dashed = zero line. Green dashed = upper part of the noise range (median of peak amplitudes = $0.5nr$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

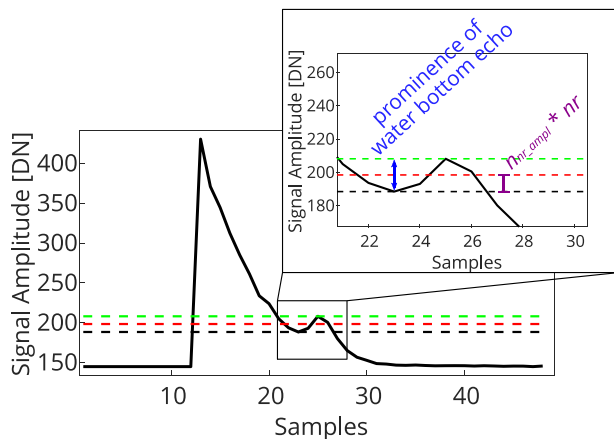
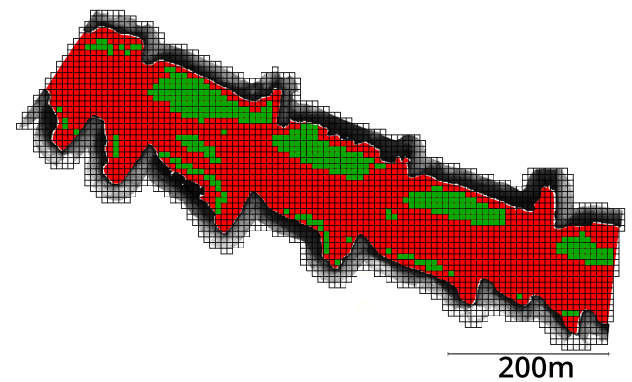


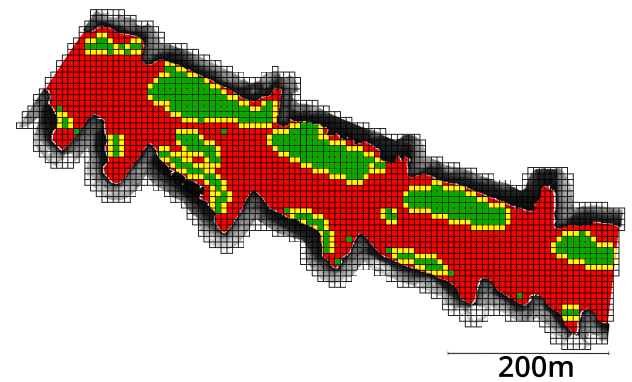
Fig. 15. Application of multiple noise ranges (nr_{ampl} =factor) for checking the reliability of the water bottom echo. Black dashed line = amplitude of local minimum. Red dashed line = amplitude threshold ($nr_{ampl} \cdot nr$). Green dashed line = amplitude of the water bottom peak. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

derived from the tail part of the full-waveform (Fig. 13), which should not contain any echoes from objects (Guenther, 1985). The application of the last samples of a full-waveform to determine noise characteristics was described in Zhu et al. (2011) in the context of denoising.

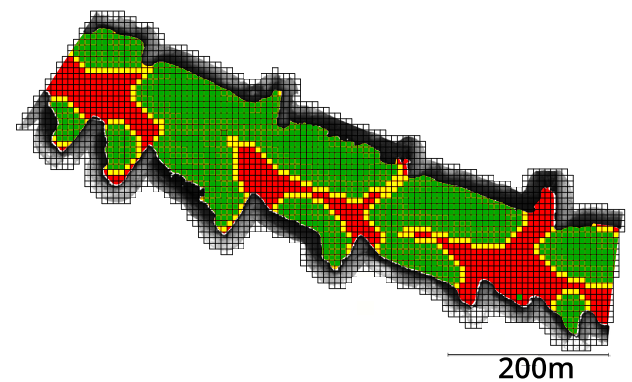
The noise range nr is determined from the median of the peak amplitudes (Fig. 14). Comparing the *prominence* of the water bottom echo with a multiple of the noise range gives an indication of the reliability of the water bottom echo found (Fig. 15). If the *prominence* is larger than $nr_{ampl} \cdot nr$, the echo can reliably be assumed to be a water bottom echo. The factor nr_{ampl} should be set in such a way



■ reliably detected water bottom echoes ■ not checked PU
(a) Iteration step 0 of 14



■ known or checked PU ■ currently checked PU ■ not yet checked PU
(b) Iteration step 1 of 14



■ known or checked PU ■ currently checked PU ■ not yet checked PU
(c) Iteration step 4 of 14

Fig. 16. Principle of iterative filtering of the processing units (PU) in the study area (for this example, 14 steps). For case (a), green = grid cells with reliably detected water bottom echoes. For cases (b) and (c), green = known or checked PU. Gray = land PU. Yellow = currently checked PU. Red = not yet checked PU. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

that, on the one hand, the reliability of the water bottom detection is maximized and, on the other hand, enough processing units are found for filtering. Fig. 16a shows all processing units with reliable detected water bottom echoes representing the starting point for the subsequent filtering process. In the first iteration step, all direct neighbors of the start processing units are determined (Fig. 16b, yellow) and their water

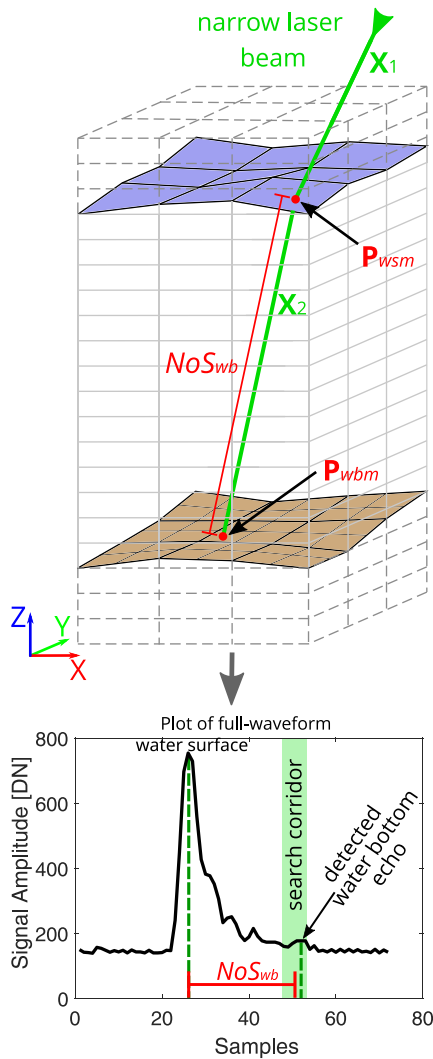


Fig. 19. Detection of weak water bottom echoes in individual full-waveforms by using ortho full-waveform analysis results (water bottom model).

4.1.1. Lateral voxel size

The lateral size plays a major role in the number of full-waveforms included in a processing unit. If the lateral size is too small, there will be less redundancy, and the signal-to-noise ratio may be poor. Weak water bottom peaks thus become more difficult to detect. If the lateral size is too large, there is a risk of introducing undesirable smoothing effects similar to a lowpass filter. However, this also depends on the water bottom topography. Table 3 summarizes the number of extracted points, mean height difference ($\overline{\Delta h}$), accuracy (RMS), and reliability (inlier rate with $|\Delta h| \leq 0.25$ m). As the voxel size increases, the number of extracted points decreases. The accuracy values improve, but the change is marginal beyond a lateral size of $2\text{ m} \times 2\text{ m}$. Even in the areas with minimum point density (cf. Fig. 2), a lateral voxel size of $1\text{ m} \times 1\text{ m}$ provides sufficient redundancy. Thus, $2\text{ m} \times 2\text{ m}$ is applied for the lateral voxel size, as more points were extracted at a negligible loss in precision and reliability. The lateral voxel size depends on scanner parameters, water turbidity and bottom topography and may thus be different for other datasets.

4.1.2. Vertical voxel size

The vertical voxel size depends on the sampling interval of the ALB system. If the vertical voxel size is too small, this can lead to very weakly occupied or even empty voxels in the processing unit. If the

Table 3

Results of the investigation of different lateral voxel sizes. The vertical voxel size was 10 cm. Only VFWSP points that could be compared with OWP and ES points were considered.

Lateral size	Extracted points	$\overline{\Delta h}$	RMS	$ \Delta h \leq 0.25\text{ m}$
$1\text{ m} \times 1\text{ m}$	1,210,441	2.1 cm	0.14 m	95.54 %
$2\text{ m} \times 2\text{ m}$	1,173,356	1.3 cm	0.12 m	96.63 %
$3\text{ m} \times 3\text{ m}$	1,138,589	0.8 cm	0.11 m	96.67 %
$4\text{ m} \times 4\text{ m}$	1,060,344	0.2 cm	0.11 m	96.34 %

Table 4

Results of the investigation of different vertical voxel sizes. The lateral voxel size applied for this was $2\text{ m} \times 2\text{ m}$. Only VFWSP points that could be compared with OWP and ES points were considered.

Vertical size	Extracted points	$\overline{\Delta h}$	RMS	$ \Delta h \leq 0.25\text{ m}$
0.065 m	1,156,801	5.5 cm	0.21 m	90.91 %
0.08 m	1,215,089	2.7 cm	0.17 m	94.50 %
0.10 m	1,173,356	1.3 cm	0.12 m	96.63 %
0.12 m	1,118,190	2.4 cm	0.12 m	96.64 %
0.14 m	1,016,103	2.3 cm	0.10 m	96.71 %

voxels are only filled with a few entries, the redundancy required to eliminate the noise signal is missing. Filling empty voxels by interpolation does not provide any additional information. If the vertical voxel size is set too large, several consecutive samples will fall into the same voxel, and smoothing effects may occur in the same way as with too large lateral voxel sizes. This further attenuates weak signals or even make them disappear.

To investigate the influence of the vertical size, the test area was processed with different voxel sizes, starting with 6.5 cm, which corresponds to the maximum vertical distance between two samples of an individual full-waveform underwater with slight deviations depending on the wave-affected incidence angle. The investigation of vertical voxel sizes smaller than 6.5 cm was omitted since no informative results regarding a suitable voxel size are expected here. Table 4 shows the results of the investigation. As the vertical voxel size increases, the number of extracted water bottom points decreases. The accuracy and reliability are lowest for vertical voxel sizes of 6.5 cm and 8.0 cm. Starting from a value of 10.0 cm, the statistical values are better and remain very similar until 14.0 cm. Since a vertical voxel size of 10.0 cm leads to the highest number of extracted points with good accuracy and reliability, this value was used for the following processing. Again, the value may be slightly different for different datasets.

4.2. Visualization of the results

Fig. 20 shows the VFWSP points color-coded by their height coordinate. Compared to the OWP points in Fig. 3, the VFWSP points cover a clearly larger area of the riverbed, resulting from an increased analyzable water depth obtained by the extended processing method. The VFWSP points seem visually plausible, as the same elevation contours can be identified as in the OWP and ES data in Figs. 3 and 4.

Fig. 21 shows an example of two cross-sections, which are aligned approximately perpendicular to the flow direction of the Elbe River (Fig. 20). For each cross-section, the VFWSP points (Fig. 21a and c) and the bottom model points (Fig. 21b and d) are shown. The VFWSP points represent the riverbed very well. Larger deviations are only evident in the near-bank area. This applies to the VFWSP points of cross-section 2 (Fig. 21c) and the bottom model points in general (Fig. 21b and d). In the cross-sections of the VFWSP results, a water depth of approximately 2.10 m (Fig. 21a) and 2.15 m (Fig. 21c) was measured. The bottom model points also represent the water bottom very well. The comparison between VFWSP and water bottom model points clearly shows that the water bottom model points are less noisy than the VFWSP points.

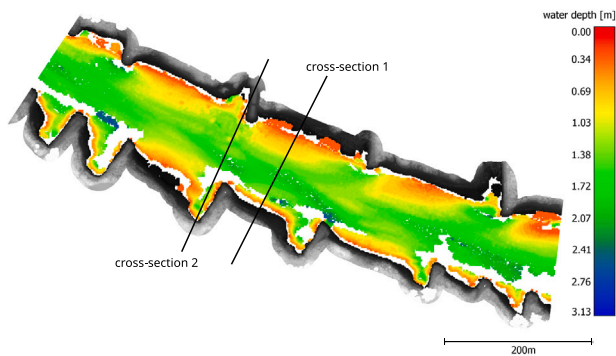


Fig. 20. VFWSP points color-coded by their water depth. Cross-section 1 is shown in Fig. 21a, b, and the cross-section 2 is shown in Fig. 21c, d. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

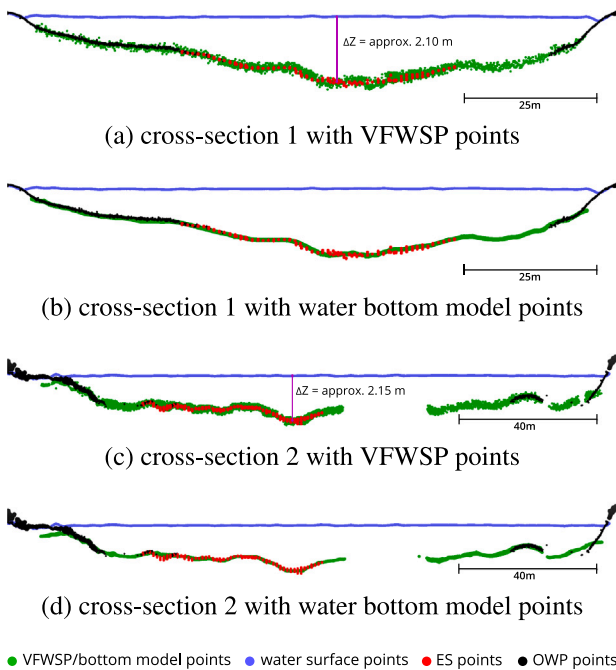


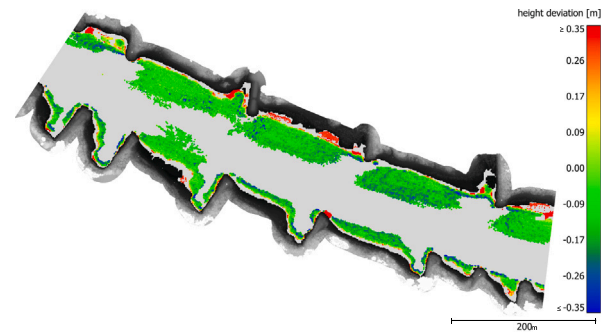
Fig. 21. Cross-sections through the Elbe River. The height coordinate is exaggerated by a factor of 6. The location of both cross-sections are shown in Fig. 20. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

It is also noticeable that in cross-section 1, the bottom of the water body could be determined completely, whereas this was not possible in cross-section 2. Since no reference data are available for this area, the actual reason remains unclear. Potential explanations are an unsteady bottom, changed reflectivity of the bottom, higher turbidity or the bottom being too deep.

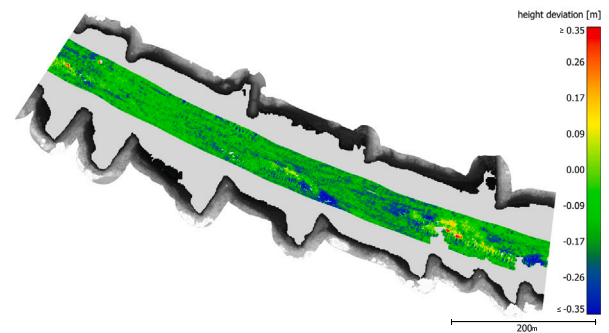
4.3. Analysis of height deviation — VFWSP points from individual full-waveforms

This subsection addresses the comparison between the resulting point cloud from the volumetric nonlinear ortho full-waveform stacking and the OWP respective ES points. Ideally, reference data with superior accuracy should be used for the analysis. The OWP and ES data, however, are considered comparative data due to their accuracy but are also called reference data in this article.

Fig. 22 shows the height deviations between VFWSP points and OWP points (Fig. 22a) as well as ES points (Fig. 22b) for the corridor



(a) Comparison VFWSP points and OWP points



(b) Comparison VFWSP points and ES points

Fig. 22. Color-coded height deviation of VFWSP points and (a) OWP points as well as (b) ES points. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

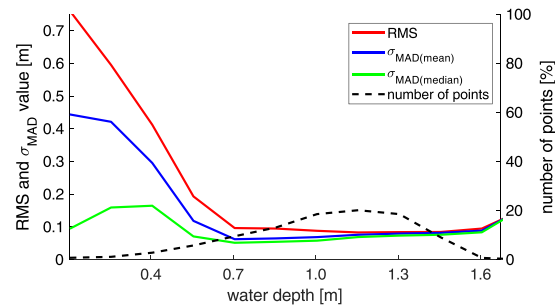
Table 5

Accuracy values and inlier rates (Section 2.4) of the comparison VFWSP points vs. OWP points (all points; points with a water depth ≤ 0.7 m; points with a water depth > 0.7 m) and VFWSP vs. ES points. wd = water depth.

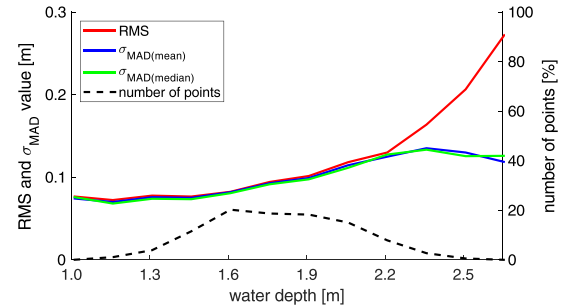
	All OWP	OWP with wd ≤ 0.7 m	OWP with wd > 0.7 m	ES
Accuracy				
$\overline{\Delta h}$	0.030 m	0.13 m	0.0094 m	-0.0020 m
$\sigma_{\overline{\Delta h}}$	0.15 m	0.27 m	0.086 m	0.10 m
RMS	0.15 m	0.30 m	0.087 m	0.10 m
$\sigma_{MAD(mean)}$	0.091 m	0.19 m	0.074 m	0.098 m
$\sigma_{MAD(median)}$	0.066 m	0.075 m	0.065 m	0.094 m
Inlier rate				
$ \Delta h \leq 15$ cm	91.01 %	77.94 %	93.76 %	87.29 %
$ \Delta h \leq 25$ cm	96.83 %	85.61 %	99.19 %	98.05 %
$ \Delta h \leq 35$ cm	98.23 %	90.63 %	99.83 %	99.63 %

with available echo-sounding data. The majority of the points have very small height differences. Higher deviations are observed in the riverbank areas, where the VFWSP points are too deep (red areas near the shore in Fig. 22a). This effect is also clearly visible in Cross-section 2 (Fig. 21c). Furthermore, the locations where the new method operates at its limits with respect to the maximum analyzable water depth show larger height deviations (blue areas in Fig. 22b). The red areas in the middle of the river result from incorrectly determined water depths in the ortho full-waveform and could not be successfully filtered by the advanced processing (Fig. 22b). The heterogeneous distribution of the height differences confirms the necessity of the water depth resolved evaluation.

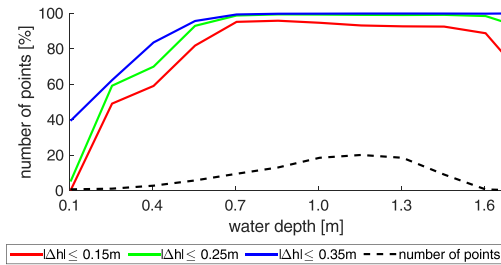
Table 5 shows the results of the comparison with a distinction of the water depth between less than and greater than 0.7 m, since investigations presented later will show that the method has problems in a range of 0.0 m to 0.7 m. OWP points are only available in the shallow-water area with a water depth between 0.0 m and 1.65 m, and



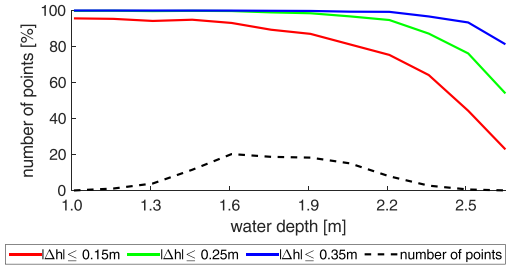
(a) Accuracy values VFWSP and OWP



(b) Accuracy values VFWSP and ES



(c) Inlier rates of VFWSP and OWP



(d) Inlier rates of VFWSP and ES

Fig. 23. Accuracy values and inlier rates for the comparison between VFWSP and OWP points as well as ES points depending on the water depth. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

the $\overline{\Delta h}$ value is positive; consequently, the VFWSP points are too deep. With the ES points, it is the opposite, where the Δh value is negative and therefore the VFWSP points are too high. However, the values are statistically insignificant according to the sigma value (significance level $\alpha = 5\%$). The RMS values of 10 cm to 15 cm or $\sigma_{MAD(median)}$ with 6.6 cm to 9.4 cm basically confirm that the extended processing method works well. This is additionally confirmed by the large values of the inlier rate of 96.83 % to 98.05 % for VFWSP points with height deviation of 25 cm or less.

In addition, a depth-dependent accuracy assessment was performed. Starting with a comparison of the VFWSP points and OWP points, Fig. 23a shows the accuracy measures, and Fig. 23c shows the reliability measures. For both curves, it is obvious that the VFWSP points with a water depth up to 0.5 m do not fit well with the OWP points. In the water depth range between 0.5 m to 0.7 m, the height differences decrease, and from 0.7 m, the VFWSP points fit the OWP points very well, which corresponds to Table 5.

The comparison of the VFWSP points with the ES points also shows very good agreement of both point clouds with respect to accuracy (Fig. 23b) and inlier rates (Fig. 23d). Up to a water depth of approximately 2.20 m, the RMS and σ_{MAD} values remain smaller than 0.15 m. With further increasing water depth, both the accuracy and inlier rate decrease noticeably. However, at this depth, the number of VFWSP points decreases rapidly; therefore, the method seems to operate at its limit.

4.4. Analysis of height deviation — VFWSP model points

Based on the ortho full-waveform results, a water bottom model was determined using the approach described in Section 3.6. This subsection discusses the question of whether this model is also suitable for representing the water bottom. For this purpose, model points with a spacing of 0.2 m were calculated, and their accuracy and reliability were tested by a comparison with OWP and ES points.

Table 6 presents the results of the investigation. Regarding the comparison of model points with OWP points, $\sigma_{MAD(mean)}$ and $\sigma_{MAD(median)}$

Table 6

Accuracy values and inlier rates (Section 2.4) of the comparison VFWSP model points vs. OWP points (all points; points with a water depth ≤ 0.7 m; points with a water depth > 0.7 m) and ES points. wd = water depth.

Accuracy	All OWP	OWP with wd ≤ 0.7 m	OWP with wd > 0.7 m	ES
$\overline{\Delta h}$	0.052 m	0.17 m	0.013 m	0.018 m
$\sigma_{\overline{\Delta h}}$	0.17 m	0.28 m	0.073 m	0.071 m
RMS	0.18 m	0.33 m	0.074 m	0.073 m
$\sigma_{MAD(mean)}$	0.096 m	0.21 m	0.055 m	0.061 m
$\sigma_{MAD(median)}$	0.055 m	0.082 m	0.050 m	0.051 m
Inlier rate				
$ \Delta h \leq 15$ cm	91.39 %	71.06 %	98.06 %	95.10 %
$ \Delta h \leq 25$ cm	95.18 %	81.85 %	99.55 %	99.00 %
$ \Delta h \leq 35$ cm	96.81 %	87.67 %	99.80 %	99.73 %

have values less than 10 cm. The inlier rates are all above 90 %. When compared to the ES points, the accuracy values are even better and are entirely in the single-digit centimeter range. The inlier rate for a threshold of 0.25 m is very good here with 99.00 %. This is confirmed by the water depth dependent evaluation illustrated in Fig. 24. As expected, the results are poor up to a water depth of 0.7 m (Table 6 column: OWP with wd ≤ 0.7 m).

The points in the remaining water depths have small height deviations validated by RMS , $\sigma_{MAD(mean)}$ and $\sigma_{MAD(median)}$ values less than 0.15 m (Fig. 24a, b) as well as high inlier rates of 86.95 % for a threshold of 0.15 m, respectively, 98.36 % for a threshold of 0.25 m in a water depth of approximately 2.20 m (Fig. 24c, d).

5. Discussion

This section first addresses the obtained results. In addition, the VFWSP results are compared to the waveform stacking method (FWS) published in Mader et al. (2021), and the pros and cons of either method are discussed. Finally, the limitations of the individual methods and possible improvements for future work are discussed.

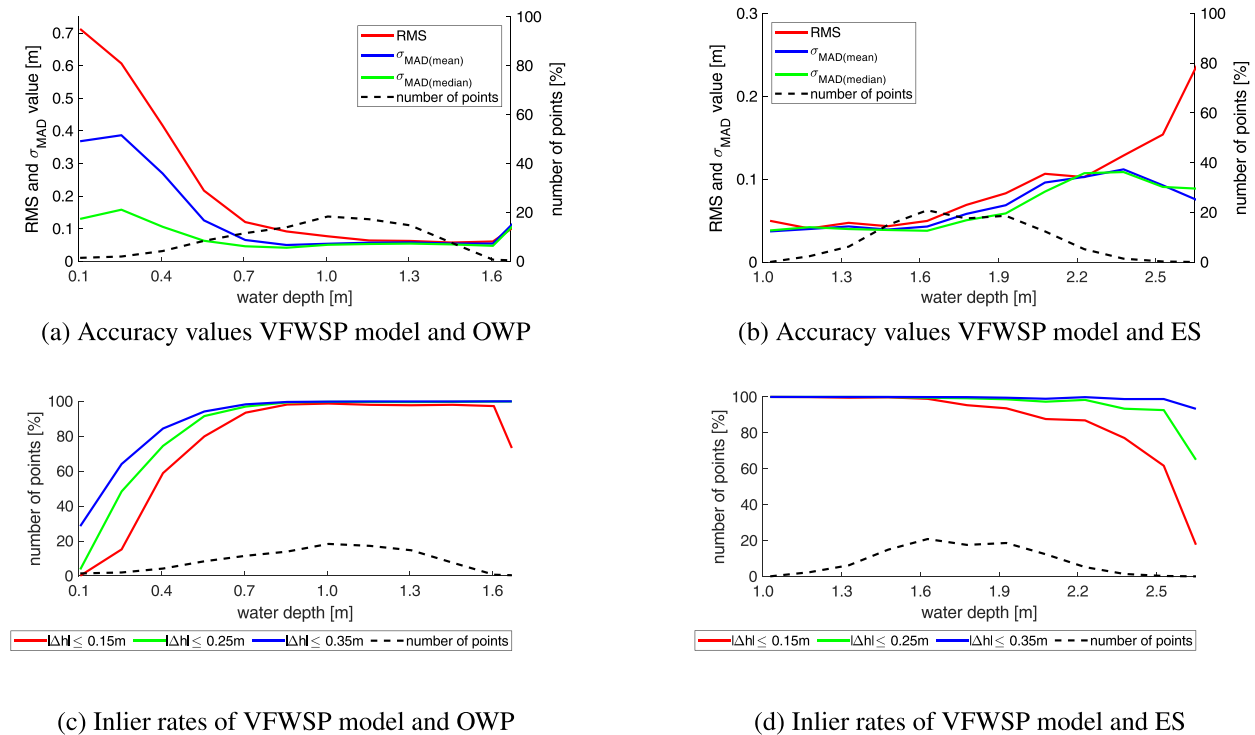


Fig. 24. Accuracy values and inlier rates for the comparison between VFWSP model points and OWP points as well as ES points depending on the water depth. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

5.1. Results

The evaluation in Table 5 and Fig. 23 shows that the volumetric nonlinear ortho full-waveform stacking worked well for the dataset at hand for water depths between 0.7 m and approximately 2.2 m. This is confirmed by the accuracy values (RMS , $\sigma_{MAD(mean)}$, $\sigma_{MAD(median)}$) smaller than 0.13 m and by the inlier rates (min. 94% at $\Delta h \leq 0.25$ m). The poor accuracy values for water depths from 0.0 m to 0.7 m were already observed in Mader et al. (2021) and explained by superimposing signals from the water surface, water column, and water bottom echo (Wagner et al., 2006; Ullrich and Pfennigbauer, 2011). In addition, the influence of the system waveform, which has a second much weaker peak after the main peak ('ringing effect'; Mallet et al., 2008), can cause problems, especially in shallow-water areas. If this second peak occurs with sufficient backscattered amplitude in the received full-waveform, there is a risk that it will be detected as a water bottom echo. However, the focus of volumetric nonlinear ortho full-waveform stacking is to increase the analyzable water depth to extract water bottom points in deeper parts of the water body. Consequently, a depth-dependent combination of OWP and VFWSP points provides the best representation of the riverbed.

5.2. Notes on full-waveform stacking methods

In Mader et al. (2021), the detection of weak bottom peaks was based on the evaluation of a stacked full-waveform, which is basically an accumulation of closely adjacent individual full-waveforms neglecting the direction of the laser beam. The neighborhood was defined purely by the horizontal position at the water surface, neglecting other properties of the employed scan mechanism (forward and backward look within a flight strip) and integrating beams from multiple, overlapping flight strips. Thus, the stacked full-waveform represents the corresponding individual full-waveforms only conditionally. For waters with low water depths, this does not play a major role. However, since this method is also applied to waters with deeper water depths (maritime waters), neglecting the laser beam direction can lead to

inaccuracies. Consequently, consideration of the laser beam pathway is necessary.

In the volumetric approach, the propagation direction of the individual laser beams is rigorously integrated into the processing pipeline, starting with voxelization (Section 3.2). The voxel structure allows the geometric consideration of slanted laser beams with different propagation directions. Thus, the characteristics of the Palmer scan pattern, the effects of combining measurement data from different strips and the propagation direction of the laser beam within the water body are considered in the voxel column and consequently also in the ortho full-waveform. In contrast to Mader et al. (2021), the lateral extent of the processing unit is the same at the water surface and at the water bottom. From a geometric point of view, this procedure should also be applicable to waters with greater depths and for data from different strips, whereby potential water surface dynamics must be considered (Section 5.5). Due to its strictly vertical orientation, the ortho full-waveform is well suited for the determination of water bottom models (Section 3.6).

Regarding the analyzable water depth (approximately 2.1 m–2.2 m), accuracy (RMS of 9–15 cm or $\sigma_{MAD(median)}$ of 6–10 cm) and reliability values (94.38%–98.05% with $|\Delta h| \leq 0.25$ m), very similar results were obtained with both methods. This shows that both stacking methods work similarly well for the studied dataset. The advantages of volumetric full-waveform stacking are expected for deeper and wavy waters; therefore, very similar results of both full-waveform stacking methods were anticipated for this dataset.

In comparison with the FWSP shown in Mader et al. (2021), the VFWSP approach shown here provides the advantage of a more regular water bottom representation with the option of nonlinearly filtered grid representation. The ortho full-waveform data also provide a basis for the derivation of water turbidity variations (Richter et al., 2021b, 2022). Voxelization and the generation of ortho full-waveforms from voxel columns eliminates the need to align full-waveforms at the water surface echoes. Thus, the water bottom model can be determined based on the ortho full-waveforms with a dynamic water surface. These advantages make the technique conditionally suitable for processing

marine coastal water airborne LiDAR bathymetry data. This means that it is possible to derive initial water depths from the ortho full-waveform as long as the water surface or separate water surfaces (from forward/backward look and overlapping flight strips; Section 5.6) are represented by a significant echo in the ortho full-waveform. If the separate water surfaces result in a disturbed water surface echo, this can be accounted for by an adjustment in the analysis of the ortho full-waveform (see Section 5.6 for details), which will be implemented in future work.

In a first pilot study with marine ALB data in the North Sea, the applicability of the two full-waveform stacking methods to coastal waters was investigated (Mader et al., 2022). In this study, the measurable water depth range could be increased by approximately 16% (to approximately 2.95 m), resulting in a significant increase in the point density and coverage of additional parts of the water bottom. The achieved *RMS* value of 0.12 m is similar to that for the Elbe dataset. A more detailed evaluation of the potential of the method for marine waters is planned in a future study.

An obvious disadvantage is the significantly larger computational effort connected with ortho full-waveform projection, which may pose practical limits when processing large datasets.

5.3. Voxel sizes

In Section 4.1, it was already stated that the voxel size has a certain influence on the quality of the results. However, the investigation in Section 4.1 shows that small variations in the voxel size have no prominent influence on the accuracy and reliability, which are very similar for lateral sizes from $2.0\text{ m} \times 2.0\text{ m}$ to $4.0\text{ m} \times 4.0\text{ m}$ and for vertical sizes from 0.10 m to 0.14 m for this dataset (Tables 3 and 4). Nevertheless, the appropriate voxel size has thus far been determined empirically, which may be rather time-consuming. Ideally, the lateral voxel size could be derived from the spatial resolution and distribution of the measured data, as well as possible prior knowledge of the water bottom topography. The waveform digitizing interval of the sensor used is certainly decisive for the vertical voxel size.

5.4. Narrow laser beam

The determination of accurate water bottom coordinates in LiDAR bathymetry requires the correction of refraction (angle and change in propagation velocity) and other propagation-induced effects (Guenther and Thomas, 1983; Richter et al., 2021a). Since the interaction of the divergent laser pulse with the wavy water surface, turbid water body, and water bottom is very complex, geometric models for the water surface, laser pulse, scattering in the water column, and diffuse reflection at the water bottom must be introduced, which are based on several simplifications. Common simplifications used in most conventional evaluation methods are (i) the hypothesis of a narrow laser beam, (ii) the neglect of scattering in the water body, and (iii) the assumption of diffuse reflection at the water bottom. This can lead to inaccuracies in water bottom determination, especially in the case of wavy water surfaces. In Westfeld et al. (2017) and Richter et al. (2018, 2021a), the effect of the simplifications on the accuracy of the water bottom coordinates was investigated and more complex models for strictly geometric modeling of laser pulse propagation were presented.

The approach presented in this article is also based on the narrow laser ray model. Since the voxels are $2\text{ m} \times 2\text{ m} \times 0.1\text{ m}$, beam divergence is only relevant for the laser pulses that pass through the voxel near the boundary to the next voxel. In such a case, strictly speaking, the amplitude of the samples is distributed between two neighboring voxels. However, since the voxel attributes are calculated from the average of all sample amplitudes in the voxel anyway, the influence of beam divergence on the shape of the ortho full-waveform is small. Nevertheless, it would in principle be possible to introduce more complex geometric models, splitting the laser beam into sub-beams. This

would result in a tremendous increase in processing time for the VFWSP approach, which would make it less practical for large-area processing. Consequently, the integration of refined models into the presented approach first requires a comprehensive analysis of the relevant signal parts of the laser footprint. Specifically it must be investigated, which parts of the expanded footprint (on the ground; Guenther, 1985) actually arrive back at the sensor and are registered. Thus, it can be assumed that the energetically very weak edge regions of the laser footprint will not play a role in the recorded full-waveform signal and thus do not have to be considered. This optimization would represent a compromise between increasing the complexity of the geometric model and acceptable processing effort.

5.5. Water surface modeling

Refraction correction requires water surface information; for instance, derived from the scanned water surface points. In the case of inland waters, it is usually acceptable to assume nearly horizontal water surfaces with only small wave movements, which justifies the use of static water surface models. This also applies to the present dataset of the Elbe River. Furthermore, dynamic water surfaces are in principle not problematic for the estimation of the initial water surface model since only the water bottom echo is searched for in the ortho full-waveform. However, for the determination and application of the search corridor, a locally valid water surface model should be used, which can be determined for every (half) strip (as e.g. in RiHydro RIEGL, 2020).

In this article, a terrain model with composite linear bivariate polynomials was used for the determination of the water surface model because the model parameters (nodal points) can be calculated quickly and effectively. This method worked well for the data in this article. However, there is a risk that undersampling surface features (waves) may lead to an inaccurate model. To cope with this, the grid cell size must be adjusted to the surface characteristics. For very smooth water surfaces, large grid cell sizes are sufficient, while rough water surfaces require smaller grid cells. Richter et al. (2021a) presented an alternative approach for modeling curved water surfaces with freeform surfaces. It allows even small details to be included in the surface model. However, modeling with freeform surfaces is very time-consuming in larger study areas, as all measurement points are included in the calculation of a water surface model point. A combined approach with the FEM method with a significantly reduced number of points for bridging smaller data gaps (e.g., few processing units) would be conceivable here.

5.6. Error influences of the water surface on the VFWSP approach

This section briefly discusses sources of error related to the water surface and their impact on VFWSP processing. The first point concerns the correct determination of the water surface point. Due to the deflection angle of the Palmer scan pattern and the property of green light penetrating water, water surface points are mostly extracted based on volume-scattering signals below the real air–water interface. A pilot study by Mandlbürger et al. (2013) investigated this effect and stated an underestimation of the water surface of 10 cm to 25 cm. According to the author, a water surface height deviation of 3 cm thereby leads to a water bottom height deviation of 1 cm. Naturally, this also affects the correct assignment of the sample amplitude of the individual full-waveforms to the voxels, which may thus be assigned to lower-lying voxels (Section 3.2). The effect of the water surface penetration error here cannot be readily determined due to the lack of sufficiently accurate reference measurements from the water bottom or near-infrared laser-scanning data from the water surface. The water surface modeling used in this article does not account for this possible error. Due to the IHO special order ($TVU = 25\text{ cm}$), this error still

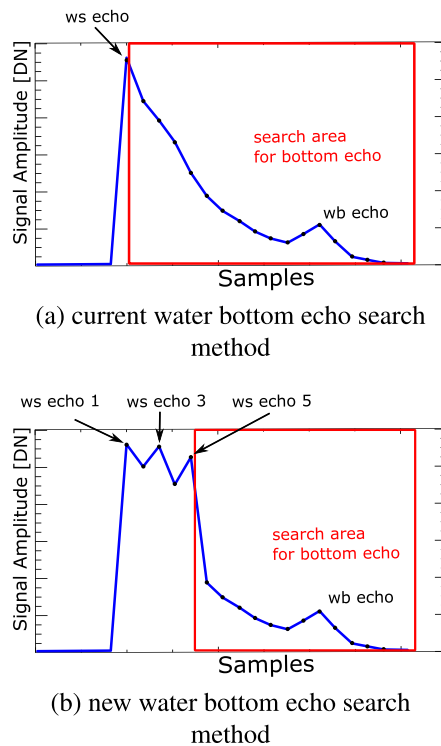


Fig. 25. Modification to account for moving and nonsteady water surfaces.

seems acceptable. However, if additional water surface information is available, a correction should be made.

The second point addresses the principle applicability of the VFWSP approach to datasets with moving or nonsteady water surfaces. For approximately horizontal water surfaces, the detection of the two most significant echoes in the ortho full-waveform (first echo = water surface; second echo = water bottom) works very well, as shown by the Elbe River dataset. For moving or nonsteady water surfaces, the application of locally valid water surface models is required to correctly map the individual full-waveform into the voxel space (Section 3.2). Consequently, separate water surface models have to be determined and applied for each (half) strip (forward/backward look in the Palmer scan) as well as for each flight strip. With respect to the ortho full-waveform, moving or nonsteady water surfaces may be represented by a superimposed or disturbed echo that cannot be analyzed with the methods presented. However, a small modification to the ortho full-waveform analysis would make this irrelevant. Currently, the search for the water bottom echo is performed in the part after the most significant peak, which is assumed to be the water surface peak (Fig. 25a). The modification is a reduction of the search area to the part of the ortho full-waveform starting after the last water surface echo whose position in the ortho full-waveform is known (Fig. 25b). This means that moving or nonsteady water surfaces have no effect on bottom detection. The advantage of this is that individual full-waveforms of forward/backward look and overlapping flight strips can still be processed together.

5.7. Water bottom model points as representation of the water bottom

To assess the suitability of the water bottom model as a representation of the water bottom, the model points were compared with the OWP and ES points in Section 4.4. The water bottom model points (not VFWSP points) used for the comparison were calculated based on the water bottom model from Section 3.6 and are uniformly distributed with a constant spacing of 0.2m in the X- and Y-directions. The analysis

showed good results; therefore, the water bottom model points are definitely a good representation for this area. The point cloud appears less noisy than the VFWSP points, and the use of the model offers the possibility to represent the water bottom in a regular point grid with rather grid spacing. In summary, the use of water bottom models has several advantages (uniform point distribution and provision of dense point clouds are possible, without gaps within a processing unit). For the present dataset with a flat water bottom, the models represent the water bottom very well. We expect that this will no longer be the case for moving terrain but cannot yet validate this due to a lack of suitable datasets.

6. Conclusion and future work

Airborne LiDAR bathymetry scans the water bottom topography with a green laser beam. The attenuation of the laser beam in medium water is significantly increased by water turbidity, thus limiting the analyzable water depth. Consequently, parts of the water bottom often cannot be detected by standard ALB processing methods. In this article, an extended processing method for full-waveform data is presented, which moves away from the isolated evaluation of full-waveform data to a joint evaluation of closely adjacent measurements: the volumetric nonlinear ortho full-waveform stacking method. The goal of the joint evaluation of closely adjacent full-waveforms is to increase the signal-to-noise ratio, i.e., to reduce or eliminate random signal components (such as noise from the sensor). Consequently, the water bottom echo is more prominent against the noise signal due to repeating echoes at the same location in the full-waveforms, thus enabling detection and evaluation of a weak water bottom peak. For the joint evaluation, local voxel space representations and the resulting ortho full-waveforms were used. Ortho full-waveforms are pseudo full-waveforms, which are generated from vertical voxel columns (voxels arranged vertically on top of each other). Based on the ortho full-waveforms, a water bottom model can be generated and used for the detection of weak water bottom echoes in individual full-waveforms. The extraction of the water bottom points from the individual full-waveforms is then a modified majority voting technique.

The results show that the volumetric nonlinear ortho full-waveform stacking approach works very well for water depths larger than 0.7 m. The validation was performed by analyzing the height deviations between VFWSP points and the online waveform processed points and respective echo sounder points. The evaluation delivered *RMS* values of 0.15 m and 0.10 m and a $\sigma_{MAD(median)}$ of 0.066 m and 0.094 m for the online waveform processed and echo sounder points. This is confirmed by good inlier rates, which state that 96.83 % to 98.05 % of the points have a maximum height deviation of 0.25 m from the reference data. Due to the poor accuracies of the VFWSP points for water depths up to approximately 0.7 m, the online waveform processed and VFWSP points should be used in a complementary manner to obtain an optimal representation of the water bottom topography. The analyzable water depth could be increased with the new method to approximately 2.20 m, which leads to an improved coverage of the riverbed of approximately 213 % (related to the coverage of the online waveform processed results).

The presented methods offer potential for further adaptation and improvement. Therefore, the following goals will be pursued in future work:

- automatic estimation of the appropriate voxel size based on the characteristics of the measured data (spatial resolution and distribution of the measured data, sensor properties, and prior knowledge of the characteristics of the water bottom topography);
- investigation of the dependence of the processing method on sensor and full-waveform characteristics;
- investigation of the influence of beam divergence on the results of full-waveform stacking processing and, based on this, development of an optimized laser beam model to account for the relevant signal components in the VFWSP (Section 5.4);

- further development of the water surface model (closing of data gaps and consideration of possible nonsteady water surfaces/dynamic water surfaces; Sections 5.5 and 5.6); and
- investigation of the potential in water bodies with deeper water depths and visual depths.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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